PES UNIVERSITY, BENGALURU

Department of Computer Science and Engineering

UE24CS341A — Software Engineering | Level 3 Mini-Project

SOFTWARE REQUIREMENTS SPECIFICATION

(SRS)

CampusReady

Online Internship & Placement Preparation Portal

Team Number	Team 12
Project Title	CampusReady – Online Internship & Placement Preparation Portal
Technology Stack	Python / JavaScript — MERN Stack (MongoDB, Express.js, React.js, Node.js)
Complexity	Medium
Stakeholders	College Training & Placement Office (TPO), participating recruiters, and students
Document Version	First Edition

Team Members

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1 Introduction and Purpose

1.1 Purpose of this Document

This Software Requirements Specification (SRS) describes, in detail, the functional and non-functional requirements for CampusReady – Online Internship & Placement Preparation Portal, developed by Team 12 as part of the UE24CS341A Software Engineering Level 3 mini-project. This document is intended to serve as the single point of reference between the development team, course instructors, and the college Training & Placement Office (TPO) for what the system will and will not do, and forms the baseline against which the system will be designed, built, tested and accepted.

1.2 Document Conventions

- Requirements are uniquely identified using the prefixes FR- (Functional Requirement), NFR- (Non-Functional Requirement), SR- (Security Requirement) and BR- (Business Rule).
- Priority levels used throughout this document are High, Medium and Low, reflecting criticality to the Sprint 1–2 delivery plan.
- The terms “portal”, “system” and “CampusReady” are used interchangeably to refer to the product being specified.

1.3 Project References

- Deliverable 1 – Synopsis / Project Proposal, Team 12, UE24CS341A.
- SRS template prescribed by the Department of Computer Science and Engineering, PES University.

2 Product Scope

Currently, students hear about internships and placements through a mix of noticeboards, generic job sites and word of mouth, and there is no single place to track an application once it has been submitted. CampusReady replaces that with one MERN-stack (MongoDB, Express.js, React.js, Node.js) web application covering internship/placement discovery, application tracking, recruiter management and placement preparation.

In scope for this mini-project:

- Student registration, authentication and profile/resume management.
- Browsing, searching and filtering of internship/job listings.
- Application submission and status tracking (Applied / Shortlisted / Rejected / Selected).
- A recruiter portal to post listings and shortlist applicants.
- A mock aptitude test and interview-preparation practice module.
- A TPO/Admin dashboard for onboarding companies, moderating postings and viewing placement analytics.
- In-app and email notifications for deadlines and status changes.

Out of scope for the current mini-project iteration: payment processing, native mobile applications, video-based interview proctoring, integration with third-party Applicant Tracking Systems (ATS), and multi-college/multi-tenant deployment. These may be considered as future enhancements beyond Sprint 2.

3 Intended Users

- Students – seeking internships/placements, building a profile and tracking applications.
- Recruiters – representing companies that post openings and shortlist candidates.
- TPO / Admin – college staff who onboard companies, moderate content and monitor outcomes.
- Guest / Visitor – an unauthenticated user who can view public information before registering.

4 Product Functions

At a high level, CampusReady provides the following major functions, each elaborated further in Sections 9 and 10:

- Account management: registration, login, password reset, and role-based access for students, recruiters and admins.
- Profile and resume building for students, including academic and skill details.
- Listing management: creation, editing, publishing and search/filtering of internship and job postings.
- Application workflow: submission, withdrawal and real-time status tracking.
- Shortlisting and recruiter-side applicant review.
- Placement preparation: mock aptitude tests and curated interview questions.
- Administrative oversight: company onboarding, posting moderation, and analytics/reporting.
- Notifications: in-app and email alerts for deadlines and status changes.

5 User Classes and Characteristics

User Class	Description	Key Privileges
Student	Registered candidate seeking internships/placements. Assumed to have basic web literacy and college-issued email/SRN.	Manage own profile; browse/apply to listings; track application status; attempt mock tests
Recruiter	Company representative onboarded by the TPO to post and manage openings.	Post/edit listings; view applicants; shortlist/reject candidates
TPO / Admin	College Training & Placement Office staff with the highest level of system access.	Onboard companies; moderate/approve postings; manage users; view analytics
Guest	Unauthenticated visitor.	View public landing pages; register for an account

6 Operating Environment

Layer	Detail
Client	Modern evergreen web browsers (Chrome, Edge, Firefox, Safari) on desktop and mobile viewports; responsive React.js single-page application
Front End	React.js (JavaScript), HTML5, CSS3
Back End	Node.js with Express.js exposing a RESTful API

Layer	Detail
Database	MongoDB (NoSQL document store)
Hosting / Deployment	Cloud or campus-hosted application server; environment-based configuration for development, staging and production
Authentication	Token-based session management (e.g., JWT) over HTTPS

7 Constraints and Assumptions

7.1 Constraints

- The project must be delivered using the Python/JavaScript MERN stack as fixed in the approved synopsis.
- Development is time-boxed to two sprints (Sprint 1–2) within a single academic semester, as per the Short-Term Plan.
- The team consists of four student developers with shared responsibility across design, development and testing.
- The system is designed for a single-college deployment and is not required to support multi-tenancy in this iteration.

7.2 Assumptions

- Students and recruiters have access to a reasonably reliable internet connection and a supported browser.
- The TPO will provide/approve the initial list of companies permitted to be onboarded as recruiters.
- Email delivery for notifications relies on a standard SMTP or transactional email service being available in the deployment environment.
- Data volumes (students, listings, applications) remain within the scale of a single college and do not require distributed database sharding.

8 External Interface Requirements

8.1 User Interfaces

- A responsive React.js web interface with distinct views/dashboards for Student, Recruiter and Admin roles.
- Consistent navigation, form validation feedback, and accessible colour contrast across all pages.

8.2 Hardware Interfaces

- None beyond a standard client device (desktop, laptop or mobile) capable of running a modern web browser.

8.3 Software Interfaces

- MongoDB database accessed via the Node.js/Express.js backend using an ODM such as Mongoose.
- SMTP/email service provider interface for outbound notification emails.
- RESTful JSON API between the React.js front end and the Express.js back end.

8.4 Communication Interfaces

- All client-server communication occurs over HTTPS.
- In-app notifications are delivered over the existing REST/WebSocket-style polling channel; email notifications are delivered via SMTP.

9 . Use Case / Analysis Models

The primary actors of CampusReady are Student, Recruiter and TPO/Admin, each interacting with the system through role-specific use cases as summarised below. Guest is a limited actor able only to register or view public content.

Use Case ID	Name	Primary Actor	Description
UC-01	Register / Login	Student, Recruiter, Admin	Create an account or authenticate to access role-specific features
UC-02	Manage Profile	Student	Create and update academic, personal and resume details
UC-03	Search & Filter Listings	Student	Browse internship/job postings using keyword, role, company or location filters
UC-04	Apply to Listing	Student	Submit an application to a chosen listing and track its status
UC-05	Post Listing	Recruiter	Create and publish a new internship/job opening
UC-06	Shortlist Applicants	Recruiter	Review submitted applications and update their status
UC-07	Take Mock Test	Student	Attempt an aptitude test or view interview-preparation content
UC-08	Onboard Company	Admin	Register a new recruiter/company on the platform
UC-09	Moderate Postings	Admin	Approve, edit or remove listings that violate platform policy
UC-10	View Analytics	Admin	View placement statistics and operational reports
UC-11	Receive Notification	Student, Recruiter	Receive in-app/email alerts for deadlines and status changes

9.1 Sample Use Case Description — UC-04 Apply to Listing

Field	Description
Actor	Student
Precondition	Student is registered, logged in, and has a complete profile
Main Flow	1. Student searches/filters listings. 2. Student opens a listing detail page. 3. Student clicks Apply. 4. System attaches the student's profile/resume to the application. 5. System sets status to Applied and notifies the student.
Alternate Flow	If the profile is incomplete, the system prompts the student to complete required fields before allowing submission.

Field	Description
Postcondition	An application record is created and visible to both the student and the owning recruiter

9.2 Use Case Diagram

Figure 1 shows the actors of CampusReady and the use cases listed above. Student, Recruiter and TPO/Admin use the system directly, while a Guest can only register. The email (SMTP) service is an external actor that delivers notifications. Applying to a listing extends searching, since a student applies from a listing found through search.

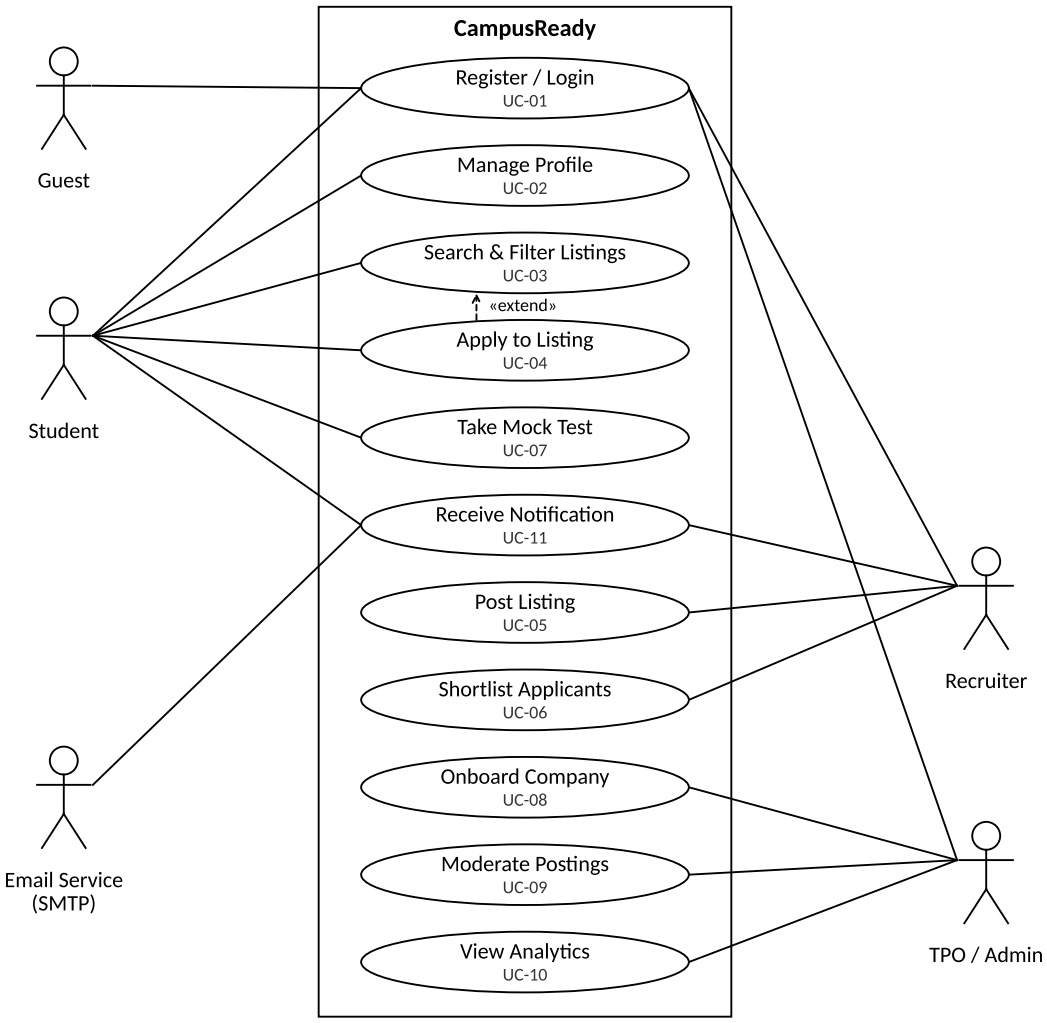


Figure 1: Use case diagram for CampusReady

10 Functional Requirements

10.1 Account & Profile Management — Owner: Sejal Siddapuram

ID	Requirement	Priority
FR-01	The system shall allow a new student to register using college-issued credentials (name, SRN/USN, email, password).	High
FR-02	The system shall authenticate registered users via login with email and password, and enforce role-based access control distinguishing Student, Recruiter and Admin capabilities.	High
FR-03	The system shall allow students to create and edit a profile including academic details, skills and an uploadable resume.	High

10.2 Listings & Applications — Owner: Sampath D Naik

ID	Requirement	Priority
FR-04	The system shall allow students to search and filter listings by keyword, role, company, location and deadline.	High
FR-05	The system shall allow a student to submit an application to a listing, attaching their current profile/resume.	High
FR-06	The system shall track and display application status as Applied, Shortlisted, Rejected or Selected, allow a student to withdraw an application prior to a recruiter decision, and notify the student when the status changes.	High

10.3 Recruiter Portal — Owner: Sagar A Walikar

ID	Requirement	Priority
FR-07	The system shall allow recruiters to create, edit and publish internship/job listings with title, description, eligibility and deadline.	High
FR-08	The system shall allow recruiters to view a list of applicants for each of their listings and update applicant status.	High
FR-09	The system shall provide recruiters with a summary count of applicants by status for each of their listings.	Medium

10.4 Admin & Preparation — Owner: Saiesh Shankar

ID	Requirement	Priority
FR-10	The system shall allow the Admin/TPO to onboard new companies/recruiter accounts and to moderate (approve, edit, or remove) postings before or after publication.	High
FR-11	The system shall provide the Admin/TPO with a dashboard summarising operational and placement statistics.	Medium
FR-12	The system shall provide a mock aptitude test module with objective questions and automatic scoring, plus curated interview-preparation content accessible to students.	Medium

11 Non-Functional Requirements

Category	Requirement
Performance	NFR-01: Core pages (listing search, dashboard) shall load within 3 seconds under normal load on a broadband connection.
Usability	NFR-02: The interface shall follow a consistent design language and be usable by a first-time student without training, on both desktop and mobile viewports.
Reliability	NFR-03: The system shall correctly persist application status transitions with no data loss during normal operation.
Availability	NFR-04: The portal shall target availability during college working hours consistent with a single-server academic deployment (best-effort outside a formal SLA).
Maintainability	NFR-05: The codebase shall follow a modular MVC-style structure (Express routes/controllers, Mongoose models, React components) to ease future enhancement.
Scalability	NFR-06: The data model and API shall be able to accommodate growth in the number of students, listings and applications for a single-college scale without redesign.
Portability	NFR-07: The front end shall render correctly on the latest two major versions of Chrome, Edge and Firefox.

12 Security Requirements

ID	Requirement
SR-01	Passwords shall be stored using a salted one-way hash (e.g., bcrypt); plaintext passwords shall never be persisted or logged.
SR-02	All client-server traffic shall be encrypted in transit using HTTPS.
SR-03	Access to Recruiter and Admin functionality shall be enforced server-side via role-based authorization checks on every protected API route, not only hidden in the UI.
SR-04	Session/authentication tokens shall have a defined expiry and be invalidated on logout.
SR-05	Uploaded resume files shall be validated for file type and size before storage to reduce the risk of malicious uploads.
SR-06	The system shall log security-relevant events (failed logins, admin actions) for audit purposes.

13 Business / Domain Rules

ID	Rule
BR-01	A student may hold at most one active application per listing; re-application requires prior withdrawal.
BR-02	Only listings approved by the Admin/TPO (directly or via pre-approved recruiter accounts) are visible to students.
BR-03	An application can only move to Shortlisted, Rejected or Selected by the recruiter who owns the corresponding listing.
BR-04	A listing past its stated deadline shall no longer accept new applications but remains visible for historical/status purposes.

ID	Rule
BR-05	Only the Admin/TPO role may onboard new recruiter/company accounts.

14 Database / Field Requirements

14.1 User (Base)

Field	Type	Constraints
_id	ObjectId	Primary key
name	String	Required
email	String	Required, unique
passwordHash	String	Required
role	Enum (Student, Recruiter, Admin)	Required
createdAt	Date	Auto-generated

14.2 Student Profile

Field	Type	Constraints
userId	ObjectId (ref: User)	Required
srn / usn	String	Required, unique
branch, semester	String / Number	Required
skills	Array<String>	Optional
resumeUrl	String	Optional

14.3 Listing

Field	Type	Constraints
_id	ObjectId	Primary key
recruiterId	ObjectId (ref: User)	Required
title, description	String	Required
eligibility	String	Optional
deadline	Date	Required
status	Enum (Draft, Published, Closed)	Required, default Draft

14.4 Application

Field	Type	Constraints
_id	ObjectId	Primary key
studentId	ObjectId (ref: User)	Required
listingId	ObjectId (ref: Listing)	Required

Field	Type	Constraints
status	Enum (Applied, Shortlisted, Rejected, Selected)	Required, default Applied
appliedAt	Date	Auto-generated

15 Glossary

Term	Definition
TPO	Training & Placement Office — the college department coordinating internships and placements
MERN Stack	MongoDB, Express.js, React.js, Node.js — the JavaScript technology stack used to build the portal
Recruiter	A company representative authorised to post and manage listings on the portal
Listing	An internship or job opening posted by a recruiter
SRN / USN	Student Registration Number / University Seat Number, the unique student identifiers used at PES University
JWT	JSON Web Token — a compact token format commonly used for stateless authentication

16 Requirement Traceability Matrix

Req ID	Source Objective / Feature	Delivered In	Verification
FR-01, FR-02, FR-03	Student registration, login, RBAC & profile management (Sejal Siddapuram)	Sprint 1–2	Working login demo, unit tests
FR-04	Listing search & filtering (Sampath D Naik)	Sprint 1	Demo + unit tests
FR-05, FR-06	Application submission, status tracking & withdrawal (Sampath D Naik)	Sprint 2	Integration tests + PRs
FR-07	Recruiter portal: post listings (Sagar A Walikar)	Sprint 1	System demo + test evidence
FR-08, FR-09	Recruiter portal: view/shortlist applicants & applicant summary (Sagar A Walikar)	Sprint 2	System demo + test evidence
FR-10, FR-11	Admin/TPO onboarding, moderation & dashboard (Saiesh Shankar)	Sprint 1–2	Validation + coverage report
FR-12	Mock aptitude test & interview-prep module (Saiesh Shankar)	Sprint 2	Validation + coverage report
SR-01 to SR-06	Security hardening	Sprint 1–2	Validation + coverage report
NFR-01 to NFR-07	Non-functional quality attributes (performance, usability, reliability, availability, maintainability, scalability, portability)	Sprint 2	Performance/load testing, usability walkthroughs, cross-browser checks, code review
BR-01 to BR-05	Business/domain rules	Sprints 1–2	Verified alongside the owning functional

Req ID	Source Objective / Feature	Delivered In	Verification
			requirement via unit/integration tests
All	System validation and release	Sprint 2	Final report + deployment/demo

17 Sprint Plan and Work Allocation

The project is delivered in two sprints, as required by the course handbook. Sprint 1 holds the features that only need the shared account module and the data model in Section 14. Sprint 2 holds the features that build on Sprint 1. Each member owns one module end to end (its features, its test cases and the related non-functional and security requirements), so members work in parallel and nobody waits on another member inside a sprint. Testing is done in the same sprint as the feature, and the final test reports are produced at the end of Sprint 2.

17.1 Sprints

Sprint	Dates	Goal
Sprint 1	1 Oct – 15 Oct 2026	Registration, login and roles; listing creation; recruiter onboarding and moderation; listing search; usable layout on desktop and mobile. Ends with the basic implementation demo.
Sprint 2	16 Oct – 6 Nov 2026	Student profile and resume; applications and status tracking; recruiter review and summary; admin dashboard; mock tests. Ends with testing, validation and the coverage reports.

Work that does not finish in Sprint 2 moves to an optional Sprint 3, as allowed by the handbook.

17.2 Work Allocation

Member	Sprint 1	Sprint 2	NFR / SR owned	Test cases
Sejal Siddapuram	FR-01, FR-02	FR-03	NFR-05; SR-01, SR-04, SR-05	TC-AUTH-01 to 04, TC-PROF-01, TC-SEC-01, TC-SEC-04, TC-SEC-05
Sampath D Naik	FR-04, NFR-02	FR-05, FR-06	NFR-02, NFR-07	TC-LIST-01, TC-APP-01 to 04, TC-NOT-01, TC-NFR-02, TC-NFR-06
Sagar A Walikar	FR-07	FR-08, FR-09	NFR-03, NFR-04; SR-02	TC-REC-01 to 04, TC-NFR-03, TC-NFR-04, TC-SEC-02, TC-SEC-07
Saiesh Shankar	FR-10	FR-11, FR-12	NFR-01, NFR-06; SR-03, SR-06	TC-ADM-01 to 03, TC-PREP-01, TC-SEC-03, TC-SEC-06, TC-NFR-01, TC-NFR-05

Every member owns three functional requirements and eight test cases. NFR-05 (maintainability) is verified by code review of every pull request rather than by a test case.

17.3 Dependencies

Task	Depends on	Available from
FR-03 Profile and resume	User and StudentProfile models (FR-01)	Sprint 1
FR-05 Apply to listing	Listing model (FR-07), approval status (FR-10), StudentProfile (FR-01)	Sprint 1

Task	Depends on	Available from
FR-06 Status and withdrawal	Application model (FR-05)	Same owner, built after FR-05
FR-08, FR-09 Recruiter review and summary	Application model (Section 14.4)	First day of Sprint 2
FR-11 Admin dashboard	Read-only counts over all models	Sprint 1, plus the Application model
FR-12 Mock test	No other module	Independent

The Application model (Section 14.4) is merged on the first day of Sprint 2, before FR-05 itself, so FR-06, FR-08, FR-09 and FR-11 can be built in parallel against it. Shared files (route registration and navigation) get one added line per feature, in the owner's own section, so merges do not conflict. Each feature is one GitHub issue and one pull request into main, tracked on the team's GitHub Projects board.